

MELISSA N. HORGER, PHD

melissahorger@gmail.com | <https://mhorger.github.io/MNHwebsite/>

TECHNICAL SKILLS

- R, Python
- Data manipulation and visualization
- Neuroimaging (pediatric EEG & MRI)
- Scientific and grant writing
- Team management
- Research presentations
- Statistical analysis
- Budget planning and management
- Industry and academic collaboration
- Mentorship and training
- Qualitative and quantitative methods
- Evidence synthesis

WORK EXPERIENCE

Postdoctoral Researcher | University of Massachusetts Amherst July 2023 - Present

- Managed a team of 15+ full time staff and undergraduate students to effectively recruit and collect physiological, behavioral, and survey data from pediatric populations and their caregivers
- Organized training opportunities for graduate students and staff in EEG methodology and multilevel modelling
- Launched 2 large federal grants, collecting longitudinal data from over 80 participants (age 9 months to 5 years)
- Spearheaded a validation study with industry partners and disseminated results to in conferences and symposia
- Lead lab-wide communications with suppliers for purchasing, troubleshooting, and updating equipment
- Served as the key decision maker and record keeper for monetary advances to compensate participants

Freelance Analyst and Tutor | UpWork Nov 2021 - Jan 2023

- Analyzed between- and within-subjects data using inferential statistics such as ANOVAs, t tests, and regression models across multiple platforms (R, SPSS)
- Generated formal and informal reports to summarize the analysis plan, results, and interpretation
- Created figures in R (ggplot2), SPSS, and Excel to visualize data (e.g., scatter plots with trend lines and confidence intervals, bar graphs, box-and-whisker plots, raincloud plots)
- Tutored individuals on how to use R for data manipulation and statistical analysis

Postdoctoral Researcher | Temple University June 2021- July 2023

- Collaborated with international companies to customize physical and digital study materials
- Wrote experimental protocols for the Institutional Review Board in compliance with ethical guidelines
- Oversaw the collection of biweekly data from 110 infants and their families across a 7 month window resulting in a repository of over 600 videos
- Performed advanced decomposition of EEG data (Principle Component Analysis; ERPs) and ran multilevel models to quantify the impact of individual-level factors on neural plasticity in the sensorimotor cortex in a sample of 60

Adjunct Instructor | Temple University & College of Staten Island Aug 2018 - May 2021

- Taught 5 sections of introductory and upper-level Psychology courses to classes of 22-49 students
- Designed semester-long group research projects with incremental goals and tangible final products
- Balanced a diverse set of student needs while maintaining high standards by writing clear and comprehensive syllabi and grading rubrics

Graduate Research Assistant | Research Foundation, CUNY Aug 2016- June 2021

- Received training in quantitative and qualitative methods and analyses through formal coursework and assignments
- Conducted literature reviews and synthesized peer reviewed research on interdisciplinary topics
- Mentored 26 undergraduates on a range of research projects, 5 of whom completed undergraduate theses
- Maintained substantial volumes of video data (>1250 hours) stored in an online repository
- Analyzed data for publication via methods such as cluster analysis, time series analysis, and logistic regressions

Educational Behavioral Support | The Vista School Sept 2015- June 2016

- Worked 1:1 with a student and trained staff on the IEP goals and responses to challenging behaviors
- Ensured fidelity to the treatment plan across staff members and provided wrap around care in the home
- Involved in planning, consultation, and progress analysis with a focus on life skills interventions

Behavior Technician | The Vista School March 2014- Sept 2015

- Set up the academic environment to elicit success from a classroom of 5-7 nonspeaking adolescents (age 9-15) with autism spectrum disorder and comorbid diagnoses
- Ran programs to meet goals established by the teacher, occupational therapist, speech/language pathologist, and physical therapist and collected data to inform progress on individual goals

- Implemented behavioral interventions for clients with a range of developmental disabilities and varying levels of care in the home, at school, and in the community
- Built rapport and maintained lines of communication between clients, families, and treatment team

GRANTS & AWARDS

- F32 Postdoctoral Individual National Research Service Award (NIH), \$157,151
- American Academy of Sleep Medicine - Young Investigators Research Forum fellow
- Postdoc Abstract Award (International Society for Developmental Psychobiology)
- Graduate Research Grant (Psi Chi), \$1,500
- Dissertation Award (CUNY), \$10,000
- Psychology Travel Award (CUNY), \$500
- Provost Pre-Dissertation Fellowship (CUNY), \$4,000
- Doctoral Student Research Grant (CUNY), \$1,400 in 2020 & \$1,500 in 2018
- Conference Presentation Support (CUNY), \$300

PRESENTATIONS

I have presented at **27** professional conferences and given **5** invited talks over the course of my career. My conference presentations encompass 31 posters and 7 symposia. Posters are succinct, data-rich visualizations, discussed 1:1 as attendees mingle. Symposia comprise 15-minute dynamic presentations with a unifying theme and discussion. Both serve as opportunities to quickly disseminate cutting-edge research and distill complex theoretical backdrops. My most recent presentations include the follow and a full list can be found on my website:

Breathing easy and sleeping soundly: Comparison of the Nanit Breathing Band to ECG sensors

- Invited speaker in the most recent symposium hosted by Nanit Labs (January, 2025)
- Synthesized and compared respiratory rates derived by Nanit wearables in combination with their proprietary computer vision algorithm to a ground truth method using a gold standard analysis (Bland-Altman plots)
- Discussed risk of bias and measurement agreement in terms accessible to a wide audience

The impact of infant sleep on current and future outcomes: Perspectives from typical and atypical populations.

- Organized a symposium for the International Congress of Infants Studies (in Glasgow, Scotland, July 2024) in which I presented with 4 other experts in the field of pediatric sleep research
- Advised on the scope of each presentation to maintain a consistent theme and iteratively build on the research
- Analyzed sleep EEG data and standardized behavioral assessments to shed light on the mechanisms through which sleep interacts with motor development, extrapolating on basic research with an applied lens

PUBLICATIONS

I have published **12** peer-reviewed articles in high quality journals including 9 empirical research papers and 3 synthesizing extant research. First authorship is indicative of my lead role in data collection, analysis, visualization, and writing. I have also served as a reviewer for 6 journals, evaluating the quality and appropriateness of submissions and offering constructive feedback in a written format.

- Horger, M. N., Campbell, K., & Marshall, P. (2024) Electrophysiological responses to digit stimulation in a tactile oddball paradigm. *International Journal of Psychophysiology*, 203, 112391.
- Horger, M. N., DeMasi, A., Allia, A. M., Scher, A., & Berger, S. E. (2023). The unique contributions of day and night sleep on infant motor problem solving. *Journal of Experimental Child Psychology*, 226, 105536.
- Horger, M.N. (2022). Novel and noninvasive methods for in-home sleep measurement and subsequent state coding in 12-month-old infants. *Infant Behavior and Development*, 69, 101775.
- Horger, M. N., Marsiliani, R. S., DeMasi, A., Allia, A., & Berger, S. E. (2021). Researcher choices for infant sleep assessment: Parent report, actigraphy, and a novel video system. *Journal of Genetic Psychology*, 182, 218-235.
- Horger, M. N. & Berger, S. E. (2019). The role of walking experience on whole-body exploration and problem solving. *Cognitive Development*, 52, 1-10.

EDUCATION

PhD, MA | Psychology | Graduate Center, CUNY

Aug 2016 - Sept 2021

BA | Psychology | Millersville University

Aug 2009 - May 2013